Unit 42 — HTB SOC Sherlock Investigation

Sysmon Log Analysis & UltraVNC Campaign

Scenario

In this Sherlock, you will familiarize yourself with Sysmon logs and various useful EventIDs for identifying and analyzing malicious activities on a Windows system. Palo Alto's Unit42 recently conducted research on an UltraVNC campaign, wherein attackers utilized a backdoored version of UltraVNC to maintain access to systems. This lab is inspired by that campaign and guides participants through the initial access stage of the campaign.

Task 1: Event ID 11 Count

- **Question:** How many Event logs are there with Event ID 11?
- **Answer:** `56`

Methodology

We filtered the Windows Event Logs for Event ID 11 and counted the total number of events.

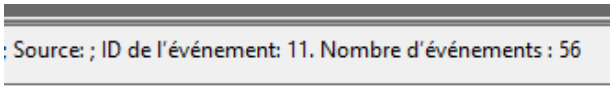


Figure 1: Event ID 11 — File Creation Events (56 total)

Conclusion

There were **56** Event ID 11 logs recorded in the system.

Task 2: Malicious Process Identification

- **Question:** What is the malicious process that infected the victim's system?
- **Answer:** `Preventivo24.02.14.exe.exe`

Methodology

Event ID 1 (Process Creation) captures every process executed on the system. This is often the first place an analyst looks when investigating an infection, as it reveals the executable name, its path, command-line arguments, and parent process. By reviewing this event, we can identify suspicious processes that deviate from normal system activity.

We filtered for Event ID 1 and examined the command lines and process paths to identify suspicious executables.

OriginalFileName	Fattura 2 2024.exe
CommandLine	"C:\Users\CyberJunkie\Downloads\Preventivo24.02.14.exe.exe"
CurrentDirectory	C:\Users\CyberJunkie\Downloads\
User	DESKTOP-887GK2L\CyberJunkie
LogonGuid	{817bddf3-311e-65cc-a7ae-1b0000000000}
LogonId	0x1baea7
TerminalSessionId	1
IntegrityLevel	Medium
Hashes	SHA1=18A24AA0AC052D31FC5B56F5C0187041174FFC61,MD5

Figure 2: Event ID 1 — Process Creation

We extracted the hash of the suspicious file and submitted it to VirusTotal:

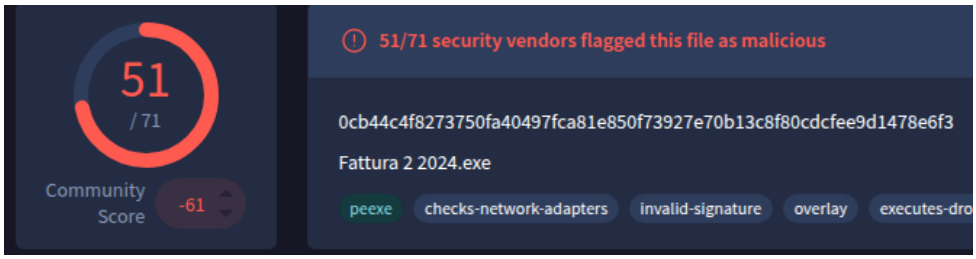


Figure 3: VirusTotal — 51/71 Detection

Conclusion

The malicious process was `Preventivo24.02.14.exe.exe`.

Task 3: Cloud Drive Distribution

- **Question:** Which Cloud drive was used to distribute the malware?
- **Answer:** `dropbox`

Methodology

When a file is downloaded from the internet, Windows creates a Zone.Identifier that stores metadata about the source of the file, including the download URL and referrer. This information is captured by Event ID 15 (FileCreateStreamHash). By analyzing this event, we can identify the original source of the malware file, which helps in threat intelligence and attribution.
We filtered for Event ID 15 related to the malware file:

TargetFilename	C:\Users\CyberJunkie\Downloads\Preventivo24.02.14.exe.exe:Zone.Identifier
CreationUtcTime	2024-02-14 03:41:26.459
Hash	SHA1=2CFE549E8DEB113DFAD2E7702637C1772ACFDBE6,MD5=41F87E73F
Contents	[ZoneTransfer] ZoneId=3 ReferrerUrl=https://www.dropbox.com/ HostUrl=https://uc2f030016253ec53f4953980a4e.dl.dropboxusercontent.com JLF54zbhziQ/file#
User	DESKTOP-887GK2L\CyberJunkie

Figure 4: Event ID 15 — Zone.Identifier showing Dropbox

Conclusion

The Zone.Identifier file revealed the file was downloaded from Dropbox.

Task 4: Time Stomping Timestamp

- Question: For many of the files it wrote to disk, the initial malicious file used a defense evasion technique called Time Stomping, where the file creation date is changed to make it appear older and blend in with other files. What was the timestamp changed to for the PDF file?
- Answer: `2024-01-14 08:10:06`

Methodology

Event ID 2 (File Creation Time Changed) tracks when a file's timestamps are modified. Attackers use Time Stomping to evade detection by making malicious files appear as if they were created long before the infection, blending in with legitimate system files.
We filtered for Event ID 2 and searched for PDF files:

RuleName	technique_id=T1070.006,technique_name=Timestomp
UtcTime	2024-02-14 03:41:58.404
ProcessGuid	{817bddf3-3684-65cc-2d02-000000001900}
ProcessId	10672
Image	C:\Users\CyberJunkie\Downloads\Preventivo24.02.14.exe.exe
TargetFilename	C:\Users\CyberJunkie\AppData\Roaming\Photo and Fax Vn\Photo and vn 1.1.2 \install\F97891C\TempFolder\~.pdf
CreationUtcTime	2024-01-14 08:10:06.029
PreviousCreationUtcTime	2024-02-14 03:41:58.404
User	DESKTOP-887GK2L\CyberJunkie

Figure 5: Event ID 2 — Time Stomping on PDF File

The attacker changed the creation time from `2024-02-14` to `2024-01-14`.

Conclusion

The PDF file was time stomped to `2024-01-14 08:10:06.029`.

Task 5: once.cmd Location

- Question: The malicious file dropped a few files on disk. Where was "once.cmd" created on disk? Please answer with the full path along with the filename.
- Answer: `C:\Users\CyberJunkie\AppData\Roaming\Photo and Fax Vn\Photo and vn 1.1.2\install\F97891C\WindowsVolume\Games\once.cmd`

Methodology

Continuing our analysis of Event ID 2 — since the malware used Time Stomping for many of the files it wrote to disk — we searched for **once.cmd** specifically.

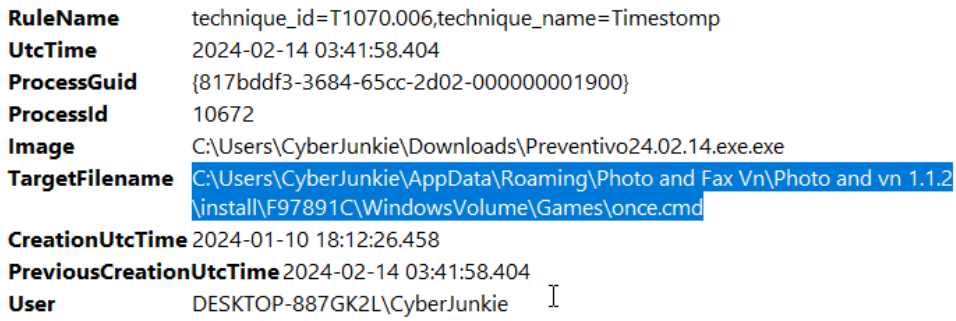


Figure 6: Event ID 2 — once.cmd Time Stomping

Conclusion

The full path of `once.cmd` is:

`C:\Users\CyberJunkie\AppData\Roaming\Photo and Fax Vn\Photo and vn 1.1.2\install\F97891C\WindowsVolume\Games\once.cmd`

Task 6: DNS Check Domain

- Question: The malicious file attempted to reach a dummy domain, most likely to check the internet connection status. What domain name did it try to connect to?
- Answer: `www.example.com`

Methodology

Event ID 22 (DNS Query) captures all DNS queries made by processes on the system. Malware often performs DNS queries to check for internet connectivity, resolve C2 domains, or exfiltrate data. By filtering this event for the malicious process, we can identify what domains the malware is attempting to reach.
We filtered for Event ID 22 related to the malicious process:

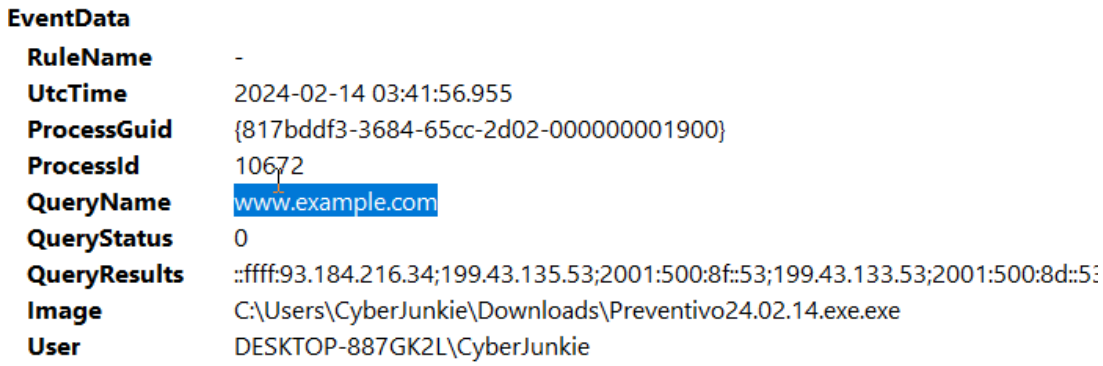


Figure 7: Event ID 22 — DNS Query to www[.]example[.]com

Conclusion

The malware attempted to resolve `www[.]example[.]com` to check internet connectivity.

Task 7: Malicious IP Address

- Question: Which IP address did the malicious process try to reach out to?
- Answer: `93[.]184[.]216[.]34`

Methodology

Event ID 3 (Network Connection) records all outbound network connections made by processes on the system. This is critical for identifying C2 (Command and Control) traffic, data exfiltration, or communication with attacker infrastructure. By filtering this event for the malicious process, we can identify the IP addresses and ports the malware communicates with.
We filtered for Event ID 3 related to the malicious process:

Image	C:\Users\CyberJunkie\Downloads\Preventivo24.02.14.exe.exe
User	DESKTOP-887GK2L\CyberJunkie
Protocol	tcp
Initiated	true
SourceIsIpv6	false
SourceIp	172.17.79.132
SourceHostname	-
SourcePort	61177
SourcePortName	-
DestinationIsIpv6	false
DestinationIp	93.184.216.34
DestinationHostname	-
DestinationPort	80

Figure 8: Event ID 3 — Network Connection

Conclusion

The malicious process connected to `93[.]184[.]216[.]34`.

Task 8: Process Termination Time

- Question: The malicious process terminated itself after infecting the PC with a backdoored variant of UltraVNC. When did the process terminate itself?
- Answer: `2024-02-14 03:41:58`

Methodology

Event ID 5 (Process Termination) captures when a process ends — whether by user action, system shutdown, or self-termination. Malware often terminates itself after completing its main objectives to avoid detection.
We filtered for Event ID 5 related to the malicious process:

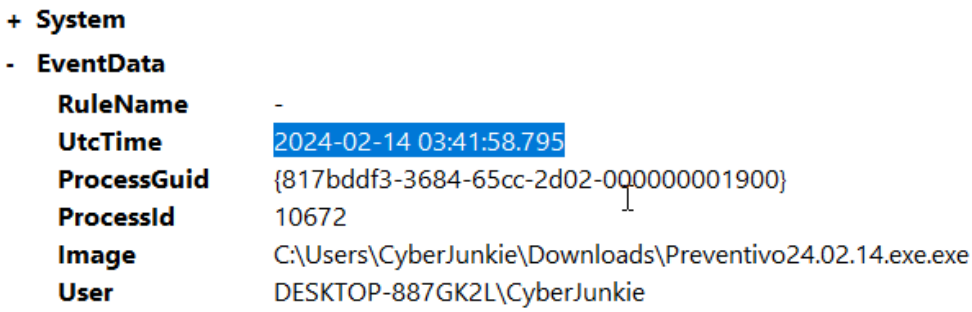


Figure 9: Event ID 5 — Process Termination

Conclusion

The malicious process terminated at `2024-02-14 03:41:58.795 UTC`.

Attack Chain Summary

The investigation successfully reconstructed the complete attack chain using Sysmon Event IDs:

1. **Initial Access:** Victim downloads `Preventivo24.02.14.exe.exe` from **Dropbox**. This was confirmed through Event ID 15 (Zone.Identifier) which captured the download source metadata.
2. **Execution:** The malicious executable is launched from the Downloads folder. Event ID 1 captured the process creation with the full path and command-line details.
3. **Defense Evasion — Time Stomping:** The malware modified file creation timestamps to evade detection, captured by Event ID 2.
4. **File Creation:** The malware dropped multiple files to disk, recorded by Event ID 11.
5. **Internet Connectivity Check:** The malware performed a DNS query for `www[.]example[.]com` to verify internet connectivity before proceeding.
6. **C2 Communication:** The malware established a network connection to `93[.]184[.]216[.]34` (port 80) — captured by Event ID 3.
7. **Self-Termination:** After completing its objectives, the malicious process terminated itself at `2024-02-14 03:41:58 UTC` to avoid detection.

Conclusion

This investigation demonstrated the effective use of Sysmon Event IDs 1, 2, 3, 5, 11, 15, and 22 for detecting and analyzing sophisticated malware campaigns. The full attack chain from initial download to process termination was successfully mapped, providing actionable intelligence for containment and remediation.